Planning and Preparation

As a VI administrator, size your environment and create a specification of the software, tools, and external services that are required to build a VCF private cloud.

VCF Planning and Preparation Workbook

The workbook is a Microsoft Excel file that helps you gather the inputs that are required for deploying a VCF fleet including workload domains, identity broker, vSphere Supervisor, and VCF Automation. You use the VCF 9.1.0 Planning and Preparation Workbook or VCF 9.1.1 Planning and Preparation Workbook for understanding the prerequisites for deploying the VCF fleet and VCF Instances, sizing your environment, as a dynamic VCF environment specification, for updating requirements according to your design, and more.

VCF Planning Tools

You can use the VMware Configurations Maximums and VMware Ports and Protocols tools to plan the VCF setup that can run your end user workloads according to the requirements of your organization.

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes. For information about how to get onboarded to VCF, see Overview of VCF Deploy, Converge, and Upgrade.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Public URLs Required for Online Functionalities for VCF and vSphere Foundation

To use online functionalities in your VCF or vSphere Foundation platform, provide access from your environment to the following list of URLs.

Purpose	Source Components	Destination URL	Port
Binaries download for components	• VCF Installer • vCenter • VCF Operations • VCF Download Tool • All VCF services runtime IP pools that host a software depot	• dl.broadcom.com • eapi.broadcom.com • vvs.broadcom.com • vsanhealth.vmware.com • vcf.packages.broadcom.com (9.1.1 and later)	443
Binaries download for Supervisor services and VCF services	• All VCF services runtime IP pools that host a software depot	• projects.packages.broadcom.com • vcf.packages.broadcom.com (9.1.1 and later)	443
Binaries download for Private AI Foundation Services	• Supervisor	• pais-docker.packages.broadcom.com	443
Telemetry data for the customer experience improvement program (CEIP)	• SDDC Manager • VCF Operations Orchestrator • All VCF services runtime IP pools	• vcsa.vmware.com	443

Purpose	Source Components	Destination URL	Port
	• vCenter • SDDC Manager • VCF Operations • VCF Operations Orchestrator • VCF Operations HCX • All VCF services runtime IP pools	• vcsa.telemetry.broadcom.com	443
	• SDDC Manager • All VCF services runtime IP pools	• scapi.telemetry.broadcom.com	443
VCF components compatibility data	• VCF Installer • SDDC Manager • VCF Download Tool	• vvs.broadcom.com	443
vSAN hardware compatibility list (HCL) data	• VCF Installer • SDDC manager • vCenter • VCF Download Tool	• vsanhealth.vmware.com • eapi.broadcom.com	443
Licensing	• VCF Operations	• vcf.broadcom.com • eapi.broadcom.com	443
Authentication for the Update Manager Download Service (UMDS)	• SDDC manager • VCF Download Tool	• auth.esp.vmware.com	443
Binary downloads Update Manager Download Service (UMDS)	• SDDC manager	• hostupdate.broadcom.com	443
Send log bundles to Broadcom by using log assist	• Cloud proxy	• eapi.broadcom.com	443
Supervisor services, image downloads, and other vSphere Kubernetes Service (VKS) functionalities	• Supervisor • vSphere Kubernetes Service (VKS) worker nodes	• packages.broadcom.com • projects.packages.broadcom.com • vcf.packages.broadcom.com (9.1.1 and later) • registry.k8s.io • projects.registry.vmware.com • registry.tkg.vmware.run	443
Download and synchronize vSphere Kubernetes Releases, OVF templates, and image publications for cluster provisioning.	• vCenter • Supervisor	• wp-content.broadcom.com	443
Observability images for VKS add-ons	• Prometheus • Alertmanager • Grafana	• quay.io	443
Policy engine images for VKS add-ons	• Kyverno	• ghcr.io	443
IDS/IPS advanced threat prevention by VMware vDefend (not part of the VCF SKU)	• NSX Manager	• api.prod.nsxti.vmware.com	443

For earlier versions of VCF and vSphere Foundation, see KB 327186.

URLs Required Only for SSL Inspection Environments

You must configure the following entries only in environments that perform deep SSL/TLS inspection. Standard firewall and proxy configurations that filter by hostname or IP and port do not require these configurations.

The Akamai entries require wildcard rules. Edge node hostnames rotate by geography and load. A specific edge hostname cannot be used reliably in place of the wildcard.

Purpose	Source Components	Destination URL	Port
AWS CDN for packages.broadcom.com	• packages.broadcom.com	• jfrog-prod-usw2-shared-oregon-main.s3.amazonaws.com	443
Akamai CDN for dl.broadcom.com	• VCF Download Tool • VCF Installer	• *.vmware.com.edgekey.net	443
Akamai CDN for eapi.broadcom.com	• VCF Download Tool • VCF Installer	• *.akamaiedge.net	443
Cloudflare CDN for wp-content.broadcom.com	• vCenter	• wp-content.broadcom.com.cdn.cloudflare.net	443

VCF Components FQDNs and IP addresses

Before you deploy VMware Cloud Foundation or vSphere Foundation, you must plan and allocate fully qualified domain names (FQDNs) and IP addresses for all VCF components. Proper DNS and IP planning ensures successful deployment and prevents address conflicts with existing infrastructure.

FQDN and IP address requirements vary depending on your deployment scenario.

- **First VCF Instance FQDNs and IP addresses.** The FQDNs and IP address requirements for the first VCF Instance in a VCF fleet, including IP address ranges and sizing guidance for the VCF management network.
- **Additional VCF Instance FQDNs and IP addresses.** The incremental FQDN and IP address requirements when deploying a subsequent VCF Instance in a VCF fleet.
- **vSphere Foundation FQDNs and IP Addresses.** The FQDN and IP address requirements specific to a vSphere Foundation platform deployment.

Dual Stack Supported VCF Components

VCF 9.1 and later supports IPv4 only and dual stack (IPv4 and IPv6) networking, but not every VCF component supports both stack configurations. VCF applies dual stack as a domain-wide setting, not a per-component choice. When you enable dual stack, all dual stack-capable components receive both IPv4 and IPv6 addresses. Components that support only IPv4 are provisioned with IPv4 addresses only.

A dual stack component supports either IPv4 only or both IPv4 and IPv6 simultaneously. In a dual stack deployment, communication uses IPv6 when both endpoints support it. Otherwise, IPv4 only.

> **Note:**
>
> For components that support dual stack networking, the FQDNs you configure must resolve to both IPv4 and IPv6 addresses.

For more information about dual stack networking and which VCF components support each IP family, see Dual Stack IP Addressing Model.

First VCF Instance FQDNs and IP addresses

VCF requires unique fully qualified domain names (FQDNs) and static IP addresses for all VCF components, along with properly configured forward (A) and reverse (PTR) DNS resolution for each. The top-level domain (TLD) label in the FQDN must contain ASCII letters only.

Each FQDN must resolve to a unique, currently unassigned IP address, and those IP addresses must not overlap with any IP ranges already reserved for VCF services runtime nodes or VCF Automation nodes. Unlike VCF Installer, which requires a strict, predefined number of FQDNs and IP addresses, Day-N components have variable requirements that depend on future deployment plans and scaling objectives.

Note:

Containerized services, such as VCF management services, VCF Automation, VCF Identity Broker, and VMware vSphere Supervisor, do not support `.local` domains. Traditional virtual appliances, such as vCenter, ESX hosts, NSX Manager, SDDC Manager, and VCF Operations, continue to operate with `.local` domains. The internal vSphere SSO domain (`vsphere.local`) continues to operate normally.

To determine on which networks the FQDNs and IP addresses for the following components must or can reside, see Network Deployment Options for VCF Components.

VCF Core Components and VCF Operations

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
vSphere	vCenter	1 FQDN	1 FQDN
	ESX	1 FQDN for each host	1 FQDN for each host
	vMotion	1 IP Address for each host	1 IP Address for each host
	Host TEP	1 IP Address for each active VDS uplink on each host *	1 IP Address for each active VDS uplink on each host *
vSAN	vSAN	1 IP Address for each host	1 IP Address for each host
NSX Manager	NSX Manager nodes	1 FQDN	3 FQDNs
	NSX Manager Cluster VIP	1 FQDN	1 FQDN
SDDC Manager	SDDC Manager	1 FQDN	1 FQDN
VCF Operations	Primary node	1 FQDN	1 FQDN
	Replica node	-	1 FQDN
	Data node **	-	1 FQDN
	Load balancer (Optional)	-	1 FQDN
	Cloud Proxy	1 FQDN	1 FQDN
	License Server	1 FQDN	1 FQDN

* Required for NSX overlay deployments. In VCF 9.1.1 and later, you can deploy a VLAN-backed VPC in TEP-less mode, which eliminates the Host TEP IP address requirement.

** VCF Operations data node FQDN is required only when deploying VCF in High Availability mode with a Medium or Large deployment size.

VCF Management Services and VCF Automation

All IP address ranges must be located on the VCF management network. If your environment does not have a dedicated VCF management network, you can deploy the components to the VM management network. See Network Deployment Options for VCF Components.

- For details on providing the IP addresses in the UI, see the following topics.
  - For new deployments, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.
  - For upgrades, see Deploy VCF Management Services and License Server as Part of VCF Upgrade to 9.1 and Upgrading to VCF Automation 9.1.

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
VCF Automation	VCF Automation	1 FQDN *	1 FQDN *
	VCF services runtime	1 FQDN *	1 FQDN *

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
	VCF Automation nodes **	• 5 IP Addresses — 3 IP addresses are assigned to the new VCF Automation 9.1 nodes. — 2 IP addresses are a buffer for automatic redeployment of failed nodes or for rolling updates in the future.	• 5 IP Addresses — 3 IP addresses are assigned to the new VCF Automation 9.1 nodes. — 2 IP addresses are a buffer for automatic redeployment of failed nodes or for rolling updates in the future.
VCF Management Services	Fleet components	1 FQDN *	1 FQDN *
	Instance components	1 FQDN *	1 FQDN *
	VCF services runtime	1 FQDN *	1 FQDN *
	VCF services runtime nodes ***	• Minimum: 12 IP addresses are required for deployment. • Recommended: 30 IP addresses can be added and are reserved for new VCF management services components or scaling out existing components. ****	• Minimum: 12 IP addresses are required for deployment. • Recommended: 30 IP addresses can be added and are reserved for new VCF management services components or scaling out existing components. ****
	Identity broker	1 FQDN *	1 FQDN *

* Do not use capital letters in the FQDN.

** By default, VCF Automation nodes are deployed on the VM management network. Alternatively, you can deploy them on a dedicated VLAN by using the fleet lifecycle API. Starting with VCF 9.1.1, when deploying VCF Automation as Day-N operation, you can select a dedicated VLAN by using the UI.

*** VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

You can specify the internal networks by using a JSON specification file for VCF. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Alternatively, you can specify the internal network by using the SDDC Manager API for VCF management services, or by using the fleet lifecycle API for VCF Automation. Starting with VCF 9.1.1 you can specify the internal networks during the deployment procedures by using the VCF Installer UI or VCF Operations UI.

**** Additional IP addresses for VCF management services are required for adding new components, scaling-out existing components, and adding new VCF services runtime nodes as an upgrade to future versions. To add new IP ranges for VCF management services components, see Network Actions for VCF Services Runtime. To scale up or scale out existing VCF management components, see Scaling VCF Management Components.

Day-N Components

The IP addresses for the log management and real-time metrics components are allocated from the IP block for VCF services runtime nodes.

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
VCF Management Services	Log management	• 1 FQDN * • 6 IP Addresses • 1 IP Address for each additional small or medium replica ** • 2 IP Addresses for each additional large replica **	• 1 FQDN * • 6 IP Addresses • 1 IP Address for each additional small or medium replica ** • 2 IP Addresses for each additional large replica **
	Real-time metrics	• 6 IP Addresses • 1 IP Address for each additional worker node ***	• 6 IP Addresses • 1 or 2 IP Addresses for each additional worker node ***
VCF Operations for Networks	Platform node	1 IP Address	1 IP Address ****
	Collector node	1 IP Address	1 IP Address

* Requires a dedicated IP address outside the IP block for VCF services runtime nodes.

* The number of IP addresses per additional replica depends on the replica size. Verify that the VCF management services cluster has sufficient CPU and memory resources to run the additional replica.

*** Additional IP addresses for real-time metrics correlate with VCF services runtime worker nodes. For an HA deployment, the number of additional IP addresses depends on the other components deployed on the VCF management services cluster.

**** Reflects the initial deployment requirement, which uses a single platform node. If you scale out VCF Operations for networks to the High Availability model, you must allocate two additional IP addresses - one for each additional platform node. For more information, see High Availability VCF Operations for Networks Model.

Add-ons

The following VCF add-ons and advanced services require FQDN and IP address assignments that you must plan for alongside the core VCF components. These add-ons run in the management domain and use IP addresses on the VCF management network or on dedicated networks with connectivity to it.

- License Hub for vDefend and Avi Load Balancer
- Security Services Platform Networking
- Requirements for Deploying Avi Load Balancer in VCF Operations
- VMware Data Services Manager Detailed Design
- Prerequisites and Best Practices for Protection and Recovery Deployment

Additional VCF Instance FQDNs and IP addresses

Additional VCF instances require unique FQDNs and static IP addresses for all VCF components, along with properly configured forward (A) and reverse (PTR) DNS resolution for each. The top-level domain (TLD) label in the FQDN must contain ASCII letters only.

Each FQDN must resolve to a unique, currently unassigned IP address, and those IP addresses must not overlap with any IP ranges already reserved for VCF services runtime nodes or VCF Automation nodes. Unlike VCF Installer, which requires a strict, predefined number of FQDNs and IP addresses, Day-N components have variable requirements that depend on future deployment plans and scaling objectives.

Note:

Containerized services, such as VCF management services, VCF Automation, VCF Identity Broker, and VMware vSphere Supervisor, do not support `.local` domains. Traditional virtual appliances, such as vCenter, ESX hosts,

NSX Manager, SDDC Manager, and VCF Operations, continue to operate with `.local` domains. The internal vSphere SSO domain (`vsphere.local`) continues to operate normally.

To determine on which networks the FQDNs and IP addresses for the following components must or can reside, see Network Deployment Options for VCF Components.

VCF Core Components and VCF Operations

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
vSphere	vCenter	1 FQDN	1 FQDN
	ESX	1 FQDN for each host	1 FQDN for each host
	vMotion	1 IP Address for each host	1 IP Address for each host
	Host TEP	1 IP Address for each active VDS uplink on each host *	1 IP Address for each active VDS uplink on each host *
vSAN	vSAN	1 IP Address for each host	1 IP Address for each host
NSX Manager	NSX Manager nodes	1 FQDN	3 FQDNs
	NSX Manager cluster VIP	1 FQDN	1 FQDN
SDDC Manager	SDDC Manager	1 FQDN	1 FQDN
VCF Operations	Cloud Proxy	1 FQDN	1 FQDN

* Required for NSX overlay deployments. In VCF 9.1.1 and later, you can deploy a VLAN-backed VPC in TEP-less mode, which eliminates the Host TEP IP address requirement.

VCF Management Services

All IP address ranges must be located on the VCF management network. If your environment does not have a dedicated VCF management network, you can deploy the components to the VM management network. See Network Deployment Options for VCF Components.

- For details on providing the IP addresses in the UI, see the following topics.
  - For new deployments, see Start a New VCF Fleet or a New VCF Instance Deployment by Using the VCF Installer Deployment Wizard.
  - For upgrades, see Deploy VCF Management Services and License Server as Part of VCF Upgrade to 9.1.

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
VCF Management Services	Instance components	1 FQDN *	1 FQDN *
	VCF services runtime	1 FQDN *	1 FQDN *
	VCF services runtime nodes **	• A minimum of 12 IP addresses are required for deployment. • Recommended: 30 IP addresses can be added and are reserved for new VCF management services components or scaling out existing components. ***	• A minimum of 12 IP addresses are required for deployment. • Recommended: 30 IP addresses can be added and are reserved for new VCF management services components or scaling out existing components. ***

* Do not use capital letters in the FQDN.

** VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.

You can specify the internal network to be used during the deployment by using a JSON specification file for VCF. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Alternatively, you can specify the internal network by using the SDDC Manager API for VCF management services. Starting with VCF 9.1.1 you can specify the internal network during the deployment procedure by using the VCF Installer UI or VCF Operations UI.

*** Additional IP addresses for VCF management services are required for adding new components, scaling-out existing components, and adding new VCF services runtime nodes as an upgrade to future versions. To add new IP ranges for VCF management services components, see Network Actions for VCF Services Runtime. To scale up or scale out existing VCF management components, see Scaling VCF Management Components.

Day-N Components

The IP addresses for the identity broker and real-time metrics components are allocated from the IP block for VCF services runtime nodes.

Component Group	Component Node / Instance	Simple Deployment FQDNs and IP Addresses	High Availability Deployment FQDNs and IP Addresses
VCF Management Services	Identity broker	1 FQDN *	1 FQDN *
	Real-time metrics	• 6 IP Addresses • 1 IP Address for each additional worker node **	• 6 IP Addresses • 1 or 2 IP Addresses for each additional worker node **

* Do not use capital letters in the FQDN. The FQDN requires a dedicated IP address outside the IP block for VCF services runtime nodes.

** Additional IP addresses for real-time metrics correlate with VCF services runtime worker nodes. For an HA deployment, the number of additional IP addresses depends on the other components deployed on the VCF management services cluster.

Add-ons

The following VCF add-ons and advanced services require FQDN and IP address assignments that you must plan for alongside the core VCF components. These add-ons run in the management domain and use IP addresses on the VCF management network or on dedicated networks with connectivity to it.

- Security Services Platform Networking
- VMware Data Services Manager Detailed Design
- Prerequisites and Best Practices for Protection and Recovery Deployment

vSphere Foundation FQDNs and IP Addresses

vSphere Foundation requires unique FQDNs and static IP addresses for all VCF components, along with properly configured forward (A) and reverse (PTR) DNS resolution for each. The top-level domain (TLD) label in the FQDN must contain ASCII letters only.

Each FQDN must resolve to a unique, currently unassigned IP address, and those IP addresses must not overlap with any IP ranges already reserved. Unlike VCF Installer, which requires a strict, predefined number of FQDNs and IP addresses, Day-N components have variable requirements that depend on future deployment plans and scaling objectives.

> **Note:**
>
> Containerized services, such as VCF management services, VCF Automation, VCF Identity Broker, and VMware vSphere Supervisor, do not support `.local` domains. Traditional virtual appliances, such as vCenter, ESX hosts, NSX Manager, SDDC Manager, and VCF Operations, continue to operate with `.local` domains. The internal vSphere SSO domain (`vsphere.local`) continues to operate normally.

To determine on which networks the FQDNs and IP addresses for the following components must or can reside, see Network Deployment Options for VCF Components.

vSphere Foundation Core Components and VCF Operations

Component Group	Component Node / Instance	FQDNs and IP Addresses
vSphere	vCenter	1 FQDN
	ESX	1 FQDN for each host
	vMotion	1 IP Address for each host
vSAN	vSAN	1 IP Address for each host
VCF Operations	Primary node	1 FQDN
	Cloud Proxy (Optional)	1 FQDN
	License Server	1 FQDN

VCF Management Services

All IP address ranges must be located on the VCF management network. If your environment does not have a dedicated VCF management network, you can deploy the components to the VM management network. See Network Deployment Options for VCF Components.

- For details on providing the IP addresses in the UI, see the following topics.
  - For new deployments, see Start a New vSphere Foundation Deployment by Using the VCF Installer Deployment Wizard.
  - For upgrades, see Deploy VCF Management Services and License Server for vSphere Foundation.

Component Group	Component Node / Instance	FQDNs and IP Addresses
VCF Management Services	Fleet components	1 FQDN *
	Instance components	1 FQDN *
	VCF services runtime	1 FQDN *
	VCF services runtime nodes **	• A minimum of 10 IP addresses are required for deployment. • Recommended: 30 IP addresses can be added and are reserved for new VCF management services components or scaling out existing components. ***

* Do not use capital letters in the FQDN.

** VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is currently in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15.

You can specify the internal network to be used during the deployment by using a JSON specification file for vSphere Foundation. See Use a JSON Specification File to Deploy VMware Cloud Foundation or vSphere Foundation. Starting with VCF 9.1.1 you can specify the internal network during the deployment procedure by using the VCF Installer UI.

*** Additional IP addresses for VCF management services are required for adding new components, scaling-out existing components, and adding new VCF services runtime nodes as an upgrade to future versions. To add new IP ranges for VCF management services components, see Network Actions for VCF Services Runtime. To scale up or scale out existing VCF management components, see Scaling VCF Management Components.

Day-N Components

The IP addresses for the log management component are allocated from the IP block for VCF services runtime nodes.

Component Group	Component Node / Instance	FQDNs and IP Addresses
VCF Management Services	Log management	• 1 FQDN • 6 IP Addresses • 1 IP Address for each additional small or medium replica * • 2 IP Addresses for each additional large replica *

* The number of IP addresses per additional replica depends on the replica size. Verify that the VCF management services cluster has sufficient CPU and memory resources to run the additional replica.

Add-ons

The following add-ons and advanced services require FQDN and IP address assignments that you must plan for alongside the core components. These add-ons use IP addresses on the VCF management network or on dedicated networks with connectivity to it.

- Requirements for Deploying Avi Load Balancer in VCF Operations

Network Deployment Options for VCF Components

VCF supports flexible network placement for its components, providing control over traffic isolation, security boundaries, and alignment with your existing infrastructure.

VM Management Network and VCF Management Network Options

Before you deploy VCF, you must determine how the VCF components are distributed across your network.

- VM management network carries traffic for core VCF management components, such as vCenter, NSX, SDDC Manager, and cloud proxy. By default, this traffic runs on the ESX management network. If simplicity is a priority, placing all VCF management components on the ESX management network is a supported option.
- With VCF 9.1 and later, VCF management network carries traffic for VCF management services, VCF Operations, license server, and VCF Automation. You can place these components on the ESX management network, on the VM Management Network, or on a dedicated network. By default, VCF management services share the VM management network. If your environment requires strict network segmentation or must meet regulatory requirements, use a dedicated VCF management network instead. See VCF Management Network Models.
- A dedicated network reduces the network incidents impact and simplifies firewall policy management and traffic monitoring. If your organization requires separation between ESX host management traffic and the management components, for example, for security, compliance, or operational clarity, direct this traffic to a dedicated network segment.

Depending on your selection for the VM management network and VCF management network options, VCF components are deployed to different deployment locations.

VM Management Network Option	VCF Management Network Option	Deploy VCF Operations and VCF Automation to a separate VDS or NSX Segment	Deployment Location
Use ESX management network	Use VM management network	No	• On the ESX Management Network: vCenter, NSX, SDDC Manager, VCF management services*, VCF Operations, cloud proxy, license server, and VCF Automation*
Use ESX management network	Use dedicated network	No	• On the ESX Management Network: vCenter, NSX, SDDC Manager, and cloud proxy • On a dedicated VCF Management Network: VCF management services*, VCF Operations, license server, and VCF Automation*
Use dedicated Network	Use dedicated network	No	• On a dedicated VM Management Network: vCenter, NSX, SDDC Manager, and cloud proxy • On a dedicated VCF Management Network: VCF management services*, VCF Operations, license server, and VCF Automation*
Use dedicated network	Use VM management network	No	• On a dedicated VM Management Network: vCenter, NSX, SDDC Manager, VCF management services*, VCF Operations, cloud proxy, license server, and VCF Automation*
Use ESX management network	Use VM management network or Use dedicated network	Yes	• On the ESX Management Network: vCenter, NSX, SDDC Manager, and cloud proxy • On a dedicated VCF Management Network: VCF management services* • On a new custom network segment or distributed port group: VCF Operations, license server, and VCF Automation*
Use dedicated network	Use VM management network or Use dedicated network	Yes	• On a dedicated network: vCenter, NSX, SDDC Manager, and cloud proxy • On the VCF Management Network: VCF management services* • On a new custom network segment: VCF Operations, license server, and VCF Automation*

* VCF services runtime uses an internal CIDR block of 198.18.0.0/15. If this subnet is in use on your network, you must change the internal CIDR block for VCF management services to 240.0.0.0/15 or 250.0.0.0/15. Different VCF services runtime instances on the same network can work with the same internal network. For example, you can have the two instances of VCF services runtime for your VCF management services and VCF Automation using the same internal network.